CSE 221 Theory Assignment 01 & 02 [HFN Section: 09 & 26]

Part 1: Iterative Time Complexity

Q1. Find the time complexity of the following codes,

a.

```
c = 0
x=30
while c<n:
x= x + c
c += 2
```

b.

```
x = 0
i = 1
while(x<n):
x = x + 2*i
i+=1
```

c.

```
x=100
for h in range(x*x):
print(" :( ")
```

d.

```
x = 0
i = 1
while(x<n):
x = x + i
i+=2
```

e.

```
int p = 0;
for (i = n / 2; i > 1; i /= 6) {
      for (j = 2; j <= n; j *= 4) {
            for (k = 0; k <= j; k *= 3) {
                  p = p + n / 2;
            }
      }
}
```

f.

```
int p = 0;
for (i = n; i > 1; i -= 1) {
      for (j = 2; j <= n; j += 1) {
            p = p + n;
            if (p > (n ^ i)) {
                  break;
            }} }
```

```
for (i = n / 2; i > 1; i /= 6) {
      for (j = 2; j <= i; j *= 4) {
            for (k = 0; k <= j; k *= 3) {
                  p = p + n / 2;
            }
      }
}
```

g. Explain why the statement, "The running time of algorithm A is at least O(n^2)" is meaningless.

Part 2: Search

[you can write pseudocode/ programmable code/ step by step instructions if you want but NOT EXPLANATION PASSAGES, ONLY LOGICAL INSTRUCTIONS]

Q1. You are given an array containing N distinct integers in a wave-like sequence. Meaning, the numbers in the beginning are in ascending order, and after a specific position, they are in descending order. For example: [1, 3, 4, 5, 9, 6, 2, -1]

a. **You have to find the maximum number of this sequence. Can you devise an efficient algorithm such that the time complexity will be less than O(N)?**
b. **Present your solution idea as a pseudocode/ python code/ flowchart/ step-by-step instructions/ logical explanation in one-two paragraphs.**
c. **Write the time complexity of your algorithm.**

Q2. You know how binary-search works. Now you are trying to implement it to do other tasks that will take you less than O(n) time to solve. Modify the binary search algorithm so that, if there are duplicates of a number in a sorted array you should be able to tell how many times a number has been duplicated? For example: [1,3,4,5,13,15,16,16,16,16,19,21,21,23], modified_bin_search(16) should return the value 4 since, "16" is present for 4 times in this array. Do that while achieving time complexity less than O(n).

Q3. For different tasks, we often need to search for things. And most of the time, we sort the dataset before performing search operations. Once, you find the following list of numbers.

[2, 5, 1.2, 6.7, 1.7, 9.3, 2.2, 7.7, 0, −4, −5.1, 2, 5, 5.2]

a. To search an element in an unsorted array, we need O(n) time using linear search. Binary search works in O(lgn) time but the array needs to be sorted beforehand which takes at least O(nlgn) time in general. Why would you then sort the array first and then perform binary search instead of just performing linear search?

b. Can you modify count sort so that it may work with negative integers as well?

c. Can you modify count sort so that it may work with the given list?

d. Say, you are working in a system where you need to sort a very big dataset. The memory available to you can barely accommodate the data. You have two options – merge sort & quick sort. Which one should you choose? Why?

e. Construct an array where quick sort fails to work in O(nlgn) time.

Part 3: Sorting

Q1. 1. Suppose you want to sort an array containing N integers. But every element of the array is the same. Write the asymptotic running times of the following sort algorithms in this case.

a. Selection sort

b. Merge sort

Q2. You are given a list of n integers where the even indices hold numbers in decreasing order and the odd indices hold numbers in increasing order. For example, this is a list of n=7 integers

Index	0	1	2	3	4	5	6
Number	23	2	19	3	7	11	5

a. Propose a linear time algorithm to sort the list.

b. Present your algorithm as a code/ pseudocode/ flowchart/ step-by-step instructions.

Part 4: Recursive Time Complexity + Divide & Conquer Algorithm

RESOURCES: I have attached some additional resources here:

- **2. Recursion time complexity HFN.pdf**
- **2.2 Recursion time complexity HFN additional.pdf**

Q1. Calculate the time complexity of the following recurrence relations. [Any method is acceptable as long as steps are shown]

A. $T(n) = 2T(n/2) + 1/n$

B. $T(n) = 625T(n/5) + n^5$

C. $T(n) = T(n/2) + T(n/5) + n$

D. $T(n) = 2T(n/4) + n^2$

Q2. Inspired by the Karatsuba algorithm, a curious CSE student Benjamin decided to modify the algorithm in his own way. For a n digit number, instead of using subproblems of size n/2, Benjamin used subproblems of size n/3. He believes with this modification, he can get a faster algorithm than Karatsuba's. So, for finding product of two n digit numbers A and B, Benjamin splitted A into 3 subproblems (A_1 , A_2 , A_3) each with n/3 digits and B into 3 subproblems (B_1 , B_2 , B_3) each with n/3 digits. Benjamin wants to write a divide and conquer algorithm for finding the product of A and B from these smaller subproblems. (Assume n is a power of 3)

A. **Write A in terms of A_1 , A_2 , A_3 and n. Write B in terms of B_1 , B_2 , B_3 and n**

B. **Calculate the product AB from your answers in (A)**

C. **Help Benjamin write the pseudocode of the divide and conquer algorithm**

D. **Calculate the time complexity of your algorithm and validate whether Benjamin's claim of getting a faster algorithm is true or not.**

Q3. Your friend gave you a binary string B (meaning each character is either 0 or 1). He wanted to find out how to calculate the maximum number of consecutive 0s in that particular string. For example,

String: 100100000111 Maximum consecutive 0s: 5

String: 1010101010101 Maximum consecutive 0s: 1

You, as an algorithm enthusiast, know that this can be solved in linear time. However, your friend asked you to propose a Divide and Conquer approach

a) Name a suitable Divide and Conquer algorithm for this task.

b) Explain how you can apply that algorithm in this scenario. Present your idea in a pseudocode/programmable code/Flowchart/step-by-step instructions.

c) Write the time complexity of your algorithm.

CSE221 Assignment Summer 24

1. [from NLH sir's quiz] A data file of 100,000 characters contains only the characters a -e and space with the frequencies indicated. **(Space has a ascii value of 32)**

Characters	**a**	**b**	**c**	**d**	**e**	**space**
Frequency (in thousands)	**45**	**13**	**12**	**16**	**9**	**5**

a. How many bits are required to save this data without any compressing (using ASCII 8 bits)?
b. Assign Fixed-Length codes (the greedy version with the highest frequent alphabet starting from code 0 then next one 1, etc) for each character to reduce the file size.

Characters	**a**	**b**	**c**	**d**	**e**	**space**
Fixed Length Code						

c. How much space (in Bits) was saved by using Fixed Length Code? (Include Table Size)
[Table size accounts for two things, table size for a One length table exam is given below]

A	001

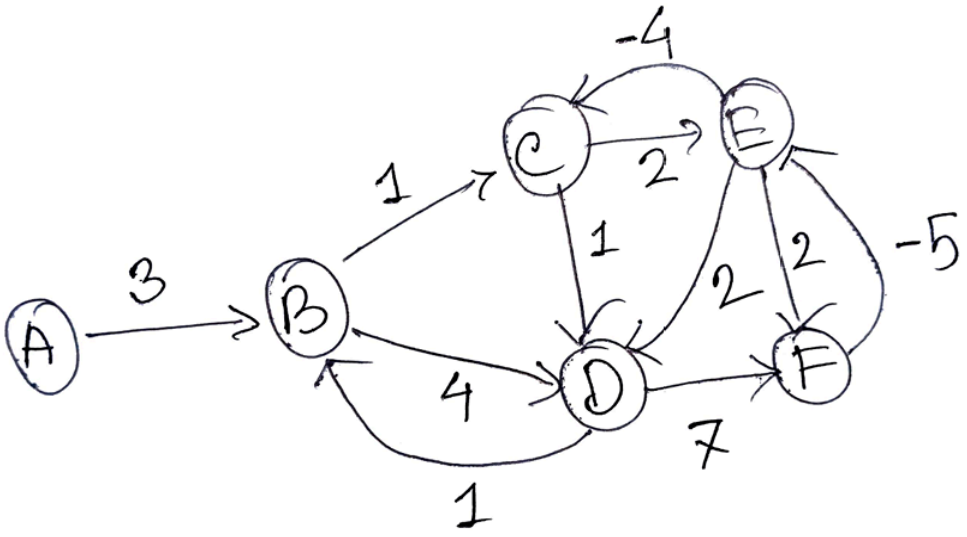
TO store the character A to 001 map, first we need A's ascii value which is of 8 bits in binary+ store the new mapped encoded value 001 meaning this table size is 8+3 = 11. Do the same for the whole table you generate]

d. Construct Huffman Tree For the given file
e. Assign huffman code to each character. [2]

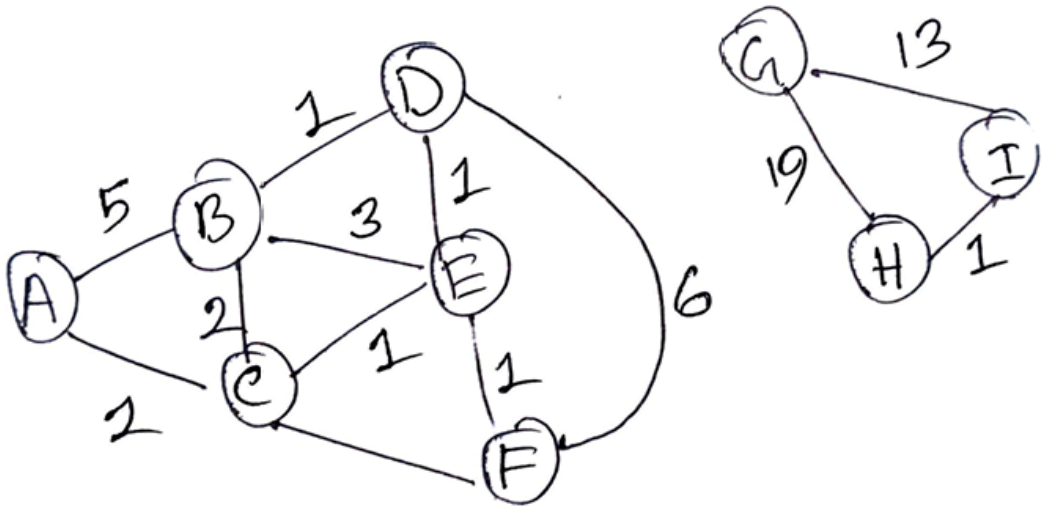
Characters	**a**	**b**	**c**	**d**	**e**	**space**
Huffman Code						

f.Encode the message "abed ec"

2. Find The Single Source Shortest Path for this graph with two algorithm
   a. **Dijkstra.** After simulation, explain whether it can solve it or not
   b. **Bellman Ford.** After simulation, explain whether it can solve it or not



3. Use two **MST algorithm** to generate the graph's MST



a. Prim's (With Source A)
b. Kruskal
c. Explain how each of the following algorithm differs in the result

4. Wye is a CSE BracU Student. The pre-advising week is approaching, and he is trying to find the most suitable course combination for next semester. As per his previous academic results, his credit limit has been set to 9. Different courses have different credits. He came up with values for thes courses by consulting with students who have already taken them. He wants to select the most valued courses.Courses, credits and values are provided below:

Course	**PHY112**	**CSE330**	**CSE331**	**CSE370**	**CSE499**
Credit	**1**	**2**	**3**	**4**	**5**
Value	**9**	**7**	**15**	**12**	**16**

a. Wye want to try all possible combinations to find the best one. Help him by simulating a suitable dynamic programming approach. Show your work in detail, then write down the selected courses.
b. Zed, a friend of Wye, adopted a slightly different approach and considered the courses based o their per-credit value. Determine whether Zed could select more valued courses than Wye. Note that, selecting a course still means taking the whole, with full credits.

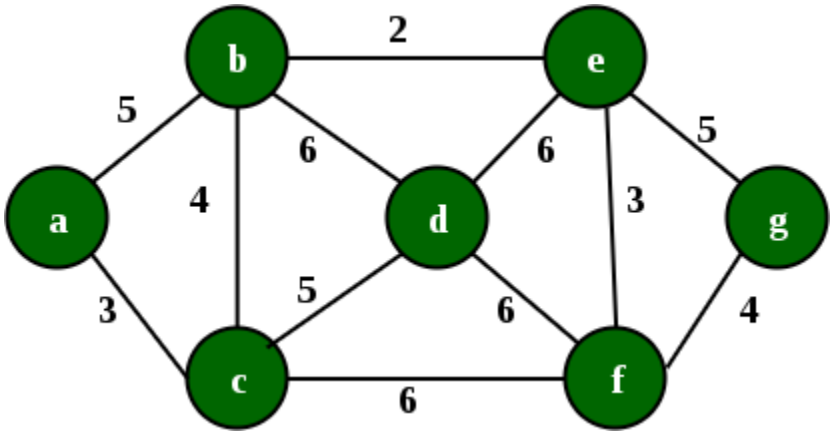
Assignment 3

1. [Mark 10]

You work as an engineer for the Roads and Highways department in a district. The district has 7 thanas that are represented by the 7 vertices in the following graph. The edges between the vertices represent the roads that connect one thana with another and the weight of an edge represents the length of the road.

A recent flood has totally damaged these roads and immediate repair work is needed. The cost to repair a road is proportionate to its length. However, your department does not have the budget to repair all the roads so you need to repair a subset of the roads. Which algorithm should you use here to find out the roads that need to be repaired to keep all the thanas connected to one another with the **minimum** possible cost?

Show the step by step simulation of the algorithm. You may pick any algorithm of your preference that you think will solve the problem.



2. [Mark 2+6+2 = 10]

Consider the following modified ASCII scheme and the encoded message using it.

Character	a	b	c	d	e	f	g	h	i	j
m_ascii	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001

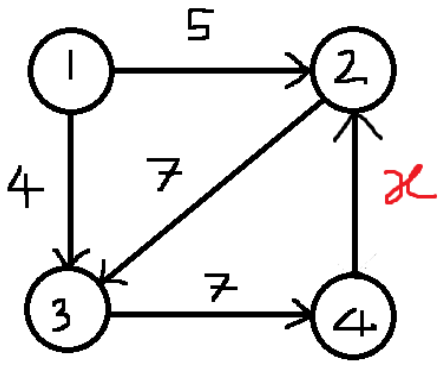
Message encoded with m_ascii code:
0000 0100 0100 1000 0100 1000 0100 1000 0001 0101 0001 0001 0110 0000 0100 1001 0010 1001

Now answer the following questions.

a) Decode the message.
b) Show simulation to find the encoding of each of the characters using Huffman coding technique. While creating a tree, less frequency nodes go to the left side of the root; left is considered as a 0, right is 1.
c) Calculate the number of bits required in the encoded message using huffman technique.

3. [Mark 4+4+3=10]

In the given graph, we want to find out the single source shortest path. The source is 1 in this case.



- Find a negative value of x such that the Dijkstra algorithm will work. Show the simulation.
- Find a negative value of x such that the Dijkstra Algorithm will not work. Use Bellman Ford Algorithm.

- Find a negative value of x such that both Dijkstra Algorithm and Bellman ford algorithm will not work. Explain the reason.

Assignment 4

4. [Mark 4+2+4=10]

Today Alice has learned how to find the Longest Common Subsequence (LCS) of two given strings. Now she wants to find the LCS of “POLYNOMIAL” and “EXPONENTIAL”.

a. Complete the LCS table and find out the length of the LCS.

index		0	1	2	3	4	5	6	7	8	9	10
		-	P	O	L	Y	N	O	M	I	A	L
0	-											
1	E											
2	X											
3	P											
4	O											
5	N											
6	E											
7	N											
8	T											
9	I											
10	A											
11	L											

b. Let M be your LCS table. What do you understand by the value of M[3][4]?
c. Find out the LCS String from the table. Show the steps of your work.
d. **[BONUS]** Write the python code for this problem.

5. [Mark 4+2+4=10]

Two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the museum are too precious, and taking a fraction of any item won't sell on the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Royal Treasury with an empty knapsack weighing a total of **7 kg** each. Even though both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Royal Treasury Museum are listed below.

Objects	Diamond	Jewelry	Sculpture	Painting	Gold Crest
Profit ($)	3	4	12	9	12
Weight (kg)	1	2	8	4	5

a) Simulate your dynamic programming algorithm to find the maximum profit Denver can make.
b) From your memory/dp table, find out which items would he take to make this amount of profit?
c) Does Nairobi's belief remain valid after the robbery? Prove it by simulating her approach.
d) **[BONUS]** Explain the DP equation for Denver's approach.
e) **[BONUS]** Write the python code for Denvers approach.

Graph Theory

Problem #1:
Lecture material:
https://drive.google.com/file/d/1OkLO91GV4M23e-1EtvFr9yeGWgk6V91q/view?usp=sharing

Some general questions regarding the basic terminology of Graph Theory

1. In a directed graph, if there is an edge from Node A to Node B, and not the other way around, can A and B be called adjacent/neighbor to each other?
2. In an undirected graph of n vertices, how many edges can we construct without allowing loops (edge to self) and multiple edges between a pair of vertices?
3. Same question for directed graphs.
4. How many edges can we construct in an undirected graph with n vertices without introducing any cycle? Loops can also be considered as cycles.
5. Same question for directed graphs.
6. Take any graph, try to construct a path that starts from a vertex, then visits all the edges of the graph exactly once. It will not be possible in all graphs, only in some having a special property. What's that property?
7. Take any graph, try to construct a path that starts from a vertex, then visits all other vertices of the graph exactly once (some edges might be unvisited). Like the previous question, it will not be possible in all graphs. Just find some examples with both cases.
8. Analyze the efficiency of using adjacency list over adjacency matrix for different type of graphs: undirected, directed, weighted, unweighted, sparse, dense…

Problem #2:
Suppose we have the following social network, Friendsbook where:

Person A is friends with Person B, C, D, and E.
Person B is friends with Person A, C, and F.
Person C is friends with Person A, B, D, and G.
Person D is friends with Person A, C, E, G, and H.
Person E is friends with Person A, D, and H.
Person F is friends with Person B and G.
Person G is friends with Person C, D, F, and H.
Person H is friends with Person D, E, and G.

A. Draw a graph to represent the network using each user as a node or vertex and the friendships as edges between the nodes.
B. Create the adjacency list and adjacency matrix for the graph representation
C. For each pair of users, calculate the number of friends they have in mutual.
D. In Friendsbook, a user can share posts. Suppose the CEO of Friendsbook hired you for developing their post display system. You were given the task to implement a system that displays posts from users who are at most two degrees of connection away from the

one who is scrolling his feed. After looking at the requirements you came up with an idea to implement the system using the BFS algorithm. Now, for person F, simulate your algorithm. Can F see the posts of all other users in his feed?

Problem #3:

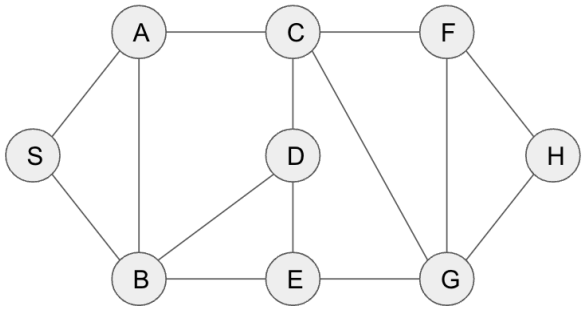
The following information is stored after implementing the DFS algorithm on an undirected graph.

Nodes	1	2	3	4	5	6	7
Parent							
Starting Time	8	11	5	6	4	1	2
Finish Time	9	12	10	7	13	14	3
Distance from Root							

A. Draw the DFS tree.
B. Fill up the empty rows of the table.

Problem #4:

Consider this graph:



A. Justify $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph.
B. Calculate the number of additional edges that can be added between the nodes randomly, keeping the graph 'simple' (without adding multiple edges between any two nodes, loops in a single node).

Assignment 1

Submission Link: https://forms.gle/vb3chAJudmMdBcjo9

Deadline: 28-11-23 (11:59PM)

Naming format: Section_ID_Name

Follow the naming format. Otherwise you will face 50% marks deduction.

There are six problems in this assignment. Each one carries the same weight.

Problem #1:

Lecture material:

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Some general questions regarding the basic terminology of Graph Theory *(Answers should be short)*

1. In a directed graph, if there is an edge from Node A to Node B, and not the other way around, can A and B be called adjacent/neighbor to each other?
2. In an undirected graph of n vertices, how many edges can we construct without allowing loops (edge to self) and multiple edges between a pair of vertices?
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A. Draw a graph to represent the network using each user as a node or vertex and the friendships as edges between the nodes.
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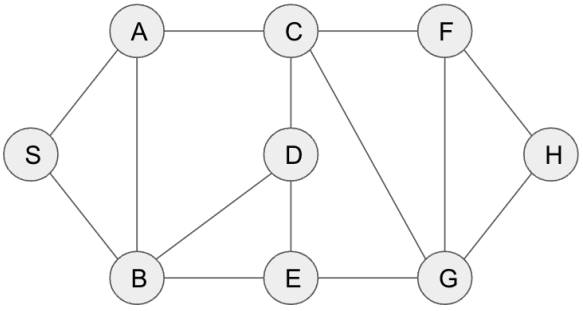
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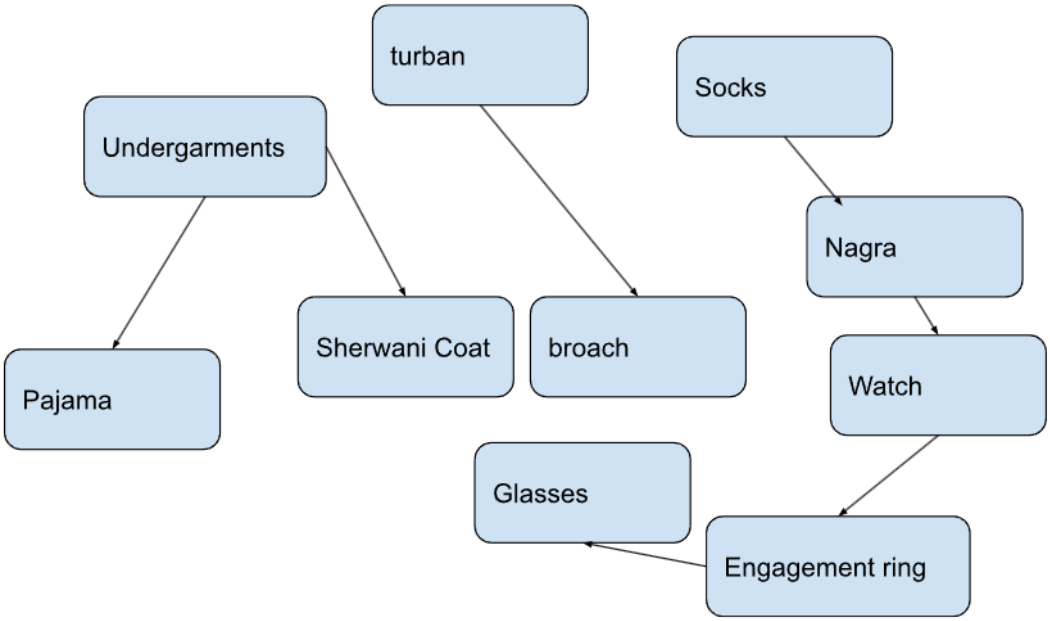


A. Justify $\sum_{v \in V} deg(v) = 2m$, where m = number of edges in the graph.

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Problem #5

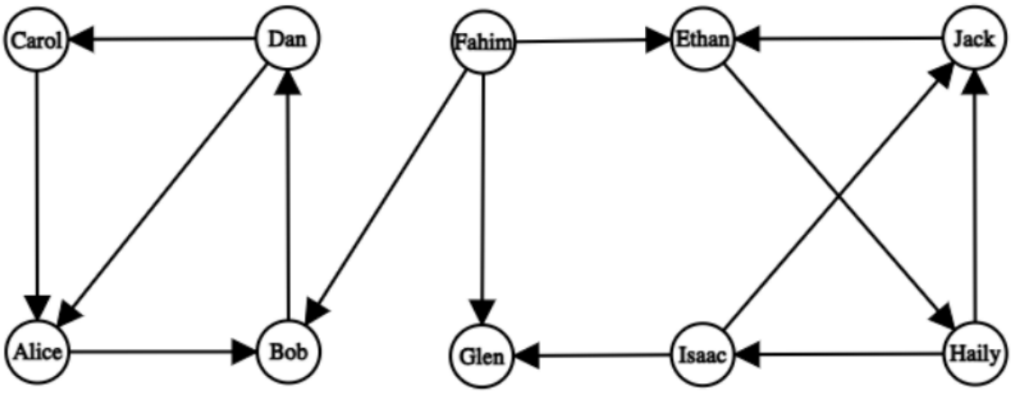
Let’s say, Anis is getting married. He came out of the shower and saw all the dresses that he needed to get dressed on the bed. Now he needs to find a system to wear them. He made up a directed Acyclic graph in his mind using the dresses. It is directed because in some cases he can not choose any order to dress. As an example, he can not wear undergarments after his pants. So in his mind, the graph looks like this:



A. Help him to find an order to put those dresses on. (Show the simulation step by step).
B. Do you think all of your classmates suggest the same ordering? Explain why you believe the answer is yes or no.

Problem #6

Among a group of 10 friends, it's not feasible for each individual to visit the houses of all the others. With an upcoming tournament in mind, the friends need to establish groups. The criterion for forming these groups is that a friend must be capable of visiting the houses of every other member in their group. The task involves calculating the count of viable groups following this criterion. A group has to consist of a minimum of two people.



a) The end result will showcase the list of groups and the members. Name the algorithm tailored for this situation.
b) Show a simulation of your chosen algorithm using the graph above. Write the number of groups that can be formed and the members of each group.
c) Those who are incapable of forming a group will be identified as individuals ineligible for tournament participation. Identify those people if there are any.
d) We also want to find the person who can visit the most houses. Propose an algorithm to implement this. Present your algorithm with a pseudocode/flowchart/step-by-step instructions.

Graph Theory II: Shortest Paths, MST

Deadline: Dec 13, 2023, 11:59pm

Assume that you are given a function called $dijkstra(G, a)$ which takes a graph G, and a source vertex a as input, then finds the shortest path (distances, $dist$ and parents, par) from a to every other vertex in the graph. Use this function to solve problems #1 and #2. Present your solution idea with a pseudocode, accompanying necessary explanation.

Problem #1: Optimum meeting place

You, and your friend Benji, live in a city which has 50 areas numbered from 1 to 50. Some of the areas are connected with bi-directional roads of various lengths. Your house is in 1 and Benji's is in 50. Both of you want to meet. Now propose an algorithm to choose the meeting point that minimizes the total travel time. In other words, if the meeting area you chose is X, it has the minimum $dist(1, X) + dist(50, X)$ among all possible area choices.
Expected time complexity: $O(E * log_2(V))$

Problem #2: Meet in the middle!
Continuing on the scenario of Problem #1, but assuming all the roads are one-way (representing a directed graph)...

Benji now has asked you to visit his house in area 50. He likes KFC, so you are thinking about getting a bucket of fried chicken for him. The are KFC outlets available in K different areas in the city. You have to make a stop at one of them. Now propose an algorithm to choose the area from the available K options that minimizes your total travel time. In other words, if the area you chose is X, it has the minimum $dist(1, X) + dist(X, 50)$ among all possible choices.
Expected time complexity: $O(E * log_2(V))$

Note: If you have to run Dijkstra's algorithm from K different sources, the complexity is $O(K * E * log_2(V))$, which exceeds the expected complexity for this problem.

Problem #3: Dijkstra's algorithm in a 'negative' weight graph!!
Draw a directed graph as described below.

No. of vertices: 7, No. of edges: 9
List of vertices, V: $\{1, 2, 3, 4, 5, 6, 7\}$
List of edges, E [in (u, v, w) format which means there is an outgoing edge from u to v with weight w] :
$\{(1, 2, 8),\ (2, 3, -8),\ (1, 3, 1)\ (3, 4, 4),\ (4, 5, -4),\ (3, 5, 1),\ (5, 6, 2),\ (6, 7, -2),\ (5, 7, 1)\}$

a. Show a simulation of Dijkstra's algorithm to find the *shortest path costs* from vertex 1 to all others. You can use the following pseudocode as reference.

   Note: Dijkstra's algorithm inserts all vertices in a priority-queue/min-heap at the beginning and then repeatedly extracts the vertex with minimum distance. It assumes, whenever a vertex is extracted from the queue, its shortest path cost from the source is finalized.

```
DIJKSTRA(G, s)
 1  for each v ∈ V[G]
 2       do d[v] ← ∞
 3          π[v] ← NIL
 4  d[s] ← 0
 5  S ← ∅
 6  Q ← V[G]
 7  while Q ≠ ∅
 8       do u ← EXTRACT-MIN(Q)
 9          S ← S ∪ {u}
10          for each vertex v ∈ Adj[u]
11               do if d[v] > d[u] + w(u, v)
12                    then d[v] ← d[u] + w(u, v)
13                         π[v] ← u
```

b. Explain why the algorithm could not find the correct shortest path costs for some of the vertices.
c. Imagine you have a modified version of Dijkstra's algorithm with the following changes:
   i. At the beginning, it inserts only the source vertex in the priority-queue.
   ii. Whenever the distance of a vertex is updated, it inserts that vertex into the priority-queue.
   iii. It does not assume that extracting a vertex from the queue means that its shortest path cost is finalized.

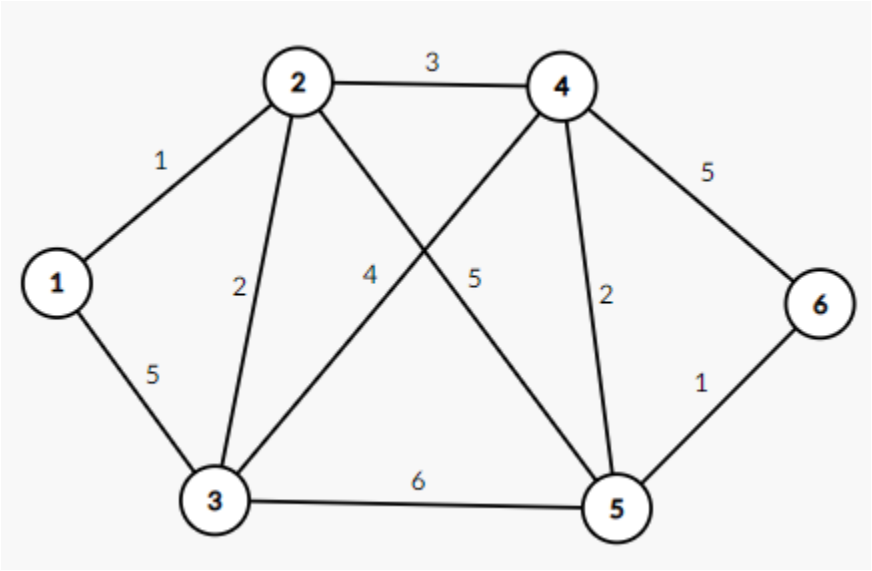
   Will this modified Dijkstra's algorithm be able to find the correct shortest path costs in the graph given above? Explain with reasoning or a simulation.
d. Analyze the time-complexity of this modified algorithm.

Problem #4: A divide-and-conquer MST algorithm?!

Your friend Mriaslun loves the divide and conquer strategy! He recently invented a divide-and-conquer algorithm to find the Minimum Spanning Tree of a given undirected weighted graph. The algorithm is quite simple, just having 4 steps in it:

1. Divide the set of vertices into two equal halves (one side can have an extra, like we do in merge sort). Assume that this divides the graph in three parts:
   a. G_{left} which contains all the vertices in the left half and the edges between them
   b. G_{right} containing the vertices in the right half, and the corresponding edges
   c. $G_{remains}$ containing the edges where one vertex is from the left half, and another one is from the right half
2. Use a recursive approach (repeating steps 1 to 4) to find the MST of G_{left}, let's call it MST_{left}
3. Do the same to find MST_{right}
4. Now you only need to select one edge from $G_{remains}$ to connect MST_{left} to MST_{right}. So you pick the one with the lowest weight value and you have the MST of the original graph!

Let's see an example.



In G_{left}, we have three vertices $\{1, 2, 3\}$ and three edges $\{(1, 2),\ (1, 3),\ (2, 3)\}$
In G_{right}, we have vertices $\{4, 5, 6\}$ and edges $\{(4, 5),\ (4, 6),\ (5, 6)\}$
And $G_{remains}$ is defined by the edges $\{(2, 4),\ (2, 5),\ (3, 4),\ (3, 5)\}$

If we can find MST_{left} and MST_{right} separately, then we can connect them with the minimum weight edge from $G_{remains}$, which is $(2, 4)$. And we can use the same divide and conquer

approach recursively to know that MST_{left} consists of the edges $\{(1,2),\ (2,3)\}$ and MST_{right} contains the edges $\{(4,5),\ (5,6)\}$.
So, the MST of the original graph is $\{(1,2),\ (2,3),\ (2,4),\ (4,5),\ (5,6)\}$ which is the correct answer!

Now your job is to prove that this algorithm will not always give a correct answer. Provide necessary reasonings to do that. You can also create an input graph and show that the algorithm fails or provides a wrong answer for this input.

Problem #5: Coin Change

There are a few types of banknotes available in your country's currency system. Any amount of taka that you want to give someone has to be constructed using these notes. Suppose you are given an infinite supply of these banknotes. You have to find the minimum number of notes needed to construct the amount.
Now, propose an algorithm applying your knowledge on graph theory which will take the currency notes and the amount to make, K as input. Then as output, return the minimum number of notes and the value of the notes.

Examples:

Input	Output
Notes: 1, 2, 5, 10 Amount to make: 21	3 notes needed: 10, 10, 1
Notes: 3, 5, 8 Amount to make: 20	4 notes needed: 5, 5, 5, 5
Notes: 3, 5, 8 Amount to make: 7	Impossible

The following questions might help:

1. Which graph theory algorithm might be relevant here?
2. How do you convert this into a graph? What are the vertices? What are the edges?

END!